Unit 2

# Electrochemistry

Chemical reactions can be made to yield electrical energy, and electrical energy can in turn drive chemical reactions that would not happen on their own.

## What this unit sets out to do

- describe an electrochemical cell; tell galvanic cells from electrolytic cells
- use the Nernst equation to find the emf of a galvanic cell; define the cell's standard potential
- derive how standard cell potential, Gibbs energy of the cell reaction and its equilibrium constant are related
- define resistivity (ρ), conductivity (κ) and molar conductivity ($\Lambda_m$) of ionic solutions
- distinguish ionic (electrolytic) conductivity from electronic conductivity
- describe how conductivity of electrolyte solutions is measured and molar conductivity calculated
- account for how conductivity and molar conductivity change with concentration; define $\Lambda_m^\circ$ (molar conductivity at zero concentration, or infinite dilution)
- state Kohlrausch law and its applications
- understand the quantitative side of electrolysis
- describe the construction of a few primary cells, secondary cells and fuel cells
- explain corrosion as an electrochemical process

## What electrochemistry covers

- making electricity from the energy of spontaneous chemical reactions
- using electrical energy to force non-spontaneous chemical changes
- matters for theory and for practice alike

## Why it matters

- electrochemical methods produce many metals and chemicals: sodium hydroxide, chlorine, fluorine and more
- batteries and fuel cells turn chemical energy into electrical energy and power a vast range of instruments and devices
- electrochemical reactions can use energy efficiently and pollute less, so the subject underpins new eco-friendly technologies

## A wide field

- signals travelling between the senses and the brain through cells, and communication between cells, are known to be electrochemical in origin
- electrochemistry is thus very broad and interdisciplinary; this unit takes up only some of its key elementary ideas

# 2.1 Electrochemical Cells

## The Daniell cell, recalled

- built from zinc and copper electrodes (Figure 2.A); it turns the chemical energy released by this redox reaction into electrical energy:

$$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$$

- its potential is 1.1 V when $Zn^{2+}$ and $Cu^{2+}$ are both at unit concentration (1 mol $dm^{-3}$)*
- such a device is a **galvanic** (or **voltaic**) cell

**Figure 2.A** Daniell cell having electrodes of zinc and copper dipping in the solutions of their respective salts.

- a zinc rod in $ZnSO_4$ solution (anode) and a copper rod in $CuSO_4$ solution (cathode)
- the two solutions are joined by a salt bridge; the rods are joined by an external wire
- electrons travel through the wire from zinc to copper, while current is taken as flowing the opposite way

## Opposing the cell with an external voltage

- apply an external potential against the cell and raise it slowly: the reaction carries on until the opposing voltage reaches 1.1 V
- at 1.1 V the reaction halts completely and no current flows
- push the external potential higher still and the reaction restarts, **in reverse**
- the cell now acts as an **electrolytic cell**: a device that uses electrical energy to bring about non-spontaneous chemical reactions; both kinds of cell are important and are examined below

**Figure 2.B** Functioning of Daniell cell when external voltage $E_{ext}$ opposing the cell potential is applied.

**(a) $E_{ext}$ < 1.1 V**

- electrons move from the Zn rod to the Cu rod, so current runs from Cu to Zn
- zinc dissolves at the anode; copper is deposited at the cathode

**(b) $E_{ext}$ = 1.1 V**

- no electron flow, no current ($I$ = 0)
- no chemical reaction takes place

**(c) $E_{ext}$ > 1.1 V**

- electrons move from Cu to Zn; current runs from Zn to Cu
- zinc is deposited on the zinc electrode; copper dissolves from the copper electrode

* Strictly, activity rather than concentration should be used. Activity is directly proportional to concentration and equals it in dilute solutions; it is treated in higher classes.

# 2.2 Galvanic Cells

## What a galvanic cell does

- a cell in which a spontaneous redox reaction releases its chemical energy as electrical energy
- the Gibbs energy of the reaction becomes electrical work, able to run a motor or appliances such as a heater, fan or geyser
- the Daniell cell is one example: $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

## Two half reactions

- added together they give the overall cell reaction:

(i) $Cu^{2+} + 2e^- \rightarrow Cu(s)$ reduction half reaction

(ii) $Zn(s) \rightarrow Zn^{2+} + 2e^-$ oxidation half reaction

- they take place in separate parts of the cell: reduction at the copper electrode, oxidation at the zinc electrode

## Half-cells

- the two parts are **half-cells** or **redox couples**: copper = reduction half-cell, zinc = oxidation half-cell
- countless galvanic cells can be built on the Daniell pattern by pairing different half-cells
- each half-cell: a metal electrode dipping into an electrolyte

## Connecting them

- outside: a metal wire through a voltmeter and a switch
- inside: the two electrolytes are linked by a **salt bridge** (Figure 2.A)
- when both electrodes stand in the same electrolyte, no salt bridge is needed

## Electrode potential

- at every electrode–electrolyte boundary, metal ions from solution tend to deposit on the electrode (making it positive), while metal atoms tend to enter solution as ions, leaving electrons behind (making it negative)
- at equilibrium the charges are separated; depending on which tendency wins, the electrode ends up positive or negative relative to the solution
- the resulting potential difference between electrode and electrolyte is the **electrode potential**
- with every species in the half-cell at unit concentration it is the **standard electrode potential**; IUPAC now uses the name standard electrode potential for what used to be called standard reduction potential

## Anode and cathode

- **anode**: the half-cell where oxidation happens; negative potential relative to the solution
- **cathode**: the half-cell where reduction happens; positive potential relative to the solution

## Flow of charge

- so a potential difference exists between the electrodes; closing the switch sends electrons from the negative to the positive electrode
- current flows the opposite way to the electrons

### Cell potential and emf

- **cell potential**: the potential difference between a galvanic cell's two electrodes, in volts
- equals the electrode (reduction) potential of the cathode minus that of the anode
- called the cell's **electromotive force (emf)** when no current is being drawn

### Writing a cell

- by accepted convention, anode on the **left**, cathode on the **right**
- single vertical line between a metal and its electrolyte; double vertical line between two electrolytes joined by a salt bridge
- written this way the emf is positive: right half-cell potential minus left

$$E_{cell} = E_{right} - E_{left}$$

### Example: copper and silver

cell reaction: $Cu(s) + 2Ag^+(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$

cathode (reduction): $2Ag^+(aq) + 2e^- \rightarrow 2Ag(s)$

anode (oxidation): $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$

- adding the two half-cell reactions gives the overall cell reaction
- silver is the cathode, copper the anode
- cell: $Cu(s)|Cu^{2+}(aq)||Ag^+(aq)|Ag(s)$
- $E_{cell} = E_{right} - E_{left} = E_{Ag^+|Ag} - E_{Cu^{2+}|Cu}$

## 2.3 Measurement of Electrode Potential

### A reference is needed

- a single half-cell's potential cannot be measured; only the difference between two, the emf, can
- fix one electrode's potential by choice, and others can be measured against it
- by convention the **standard hydrogen electrode**, $Pt(s)|H_2(g)|H^+(aq)$, is given **zero** potential at every temperature, for:

$$H^+(aq) + e^- \rightarrow \tfrac{1}{2}\, H_2(g)$$

### Building the SHE

- a platinum electrode coated with platinum black, dipped in acid solution, with pure hydrogen gas bubbled through
- both the reduced and the oxidised form of hydrogen are held at unit concentration (Figure 2.C)
- that is, hydrogen gas at one bar and hydrogen ion at one molar

**Figure 2.C** Standard Hydrogen Electrode (SHE).

- a platinum-black-coated platinum foil sits in an acidic solution of $H^+(aq)$ at 1 M, joined to the external circuit by a wire
- hydrogen gas at 1 bar enters through a surrounding tube and bubbles over the platinum surface

## Measuring against the SHE

- build a cell at 298 K with the SHE as the anode (reference half-cell) and the half-cell under test as the cathode: the emf gives the test half-cell's reduction potential
- with the oxidised and reduced forms in the right-hand half-cell at unit concentration, the cell potential equals that half-cell's standard electrode potential $E^{\ominus}_{R}$:

$$E^{\ominus} = E^{\ominus}_{R} - E^{\ominus}_{L} \;;\; E^{\ominus}_{L}\,(\text{SHE}) = 0 \;\Rightarrow\; E^{\ominus} = E^{\ominus}_{R}$$

## Copper: +0.34 V

$$\text{Pt(s)}|\text{H}_2\text{(g, 1 bar)}|\text{H}^+\text{(aq, 1 M)}||\text{Cu}^{2+}\text{(aq, 1 M)}|\text{Cu}$$

- measured emf 0.34 V, which is the standard electrode potential for $Cu^{2+}(aq, 1M) + 2e^- \rightarrow Cu(s)$
- positive: $Cu^{2+}$ is reduced more readily than $H^+$; under these standard conditions $H^+$ cannot oxidise Cu (put the other way, $H_2$ can reduce $Cu^{2+}$)
- so Cu does not dissolve in HCl; in nitric acid it is oxidised by the nitrate ion, not by $H^+$

## Zinc: −0.76 V

$$\text{Pt(s)}|\text{H}_2\text{(g, 1 bar)}|\text{H}^+\text{(aq, 1 M)}||\text{Zn}^{2+}\text{(aq, 1M)}|\text{Zn}$$

- measured emf −0.76 V, the standard electrode potential for $Zn^{2+}(aq, 1\ M) + 2e^- \rightarrow Zn(s)$
- negative: $H^+$ ions can oxidise zinc (zinc can reduce $H^+$)

## The Daniell cell by this convention

left electrode: $Zn(s) \rightarrow Zn^{2+}(aq, 1\ M) + 2e^-$
right electrode: $Cu^{2+}(aq, 1\ M) + 2e^- \rightarrow Cu(s)$
sum: $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

- emf of the cell:
  $E^{\ominus}_{cell} = E^{\ominus}_{R} - E^{\ominus}_{L} = 0.34\ \text{V} - (-0.76)\ \text{V} = \mathbf{1.10\ V}$

## Inert electrodes

- metals such as platinum or gold may serve as inert electrodes: they take no part in the reaction but offer a surface for oxidation or reduction and conduct electrons
- hydrogen electrode: $\text{Pt(s)}|\text{H}_2\text{(g)}|\text{H}^+\text{(aq)}$; half-cell reaction $H^+(aq) + e^- \rightarrow \tfrac{1}{2}\,H_2(g)$
- bromine electrode: $\text{Pt(s)}|\text{Br}_2\text{(aq)}|\text{Br}^-\text{(aq)}$; half-cell reaction $\tfrac{1}{2}\,Br_2(aq) + e^- \rightarrow Br^-(aq)$

### Reading standard electrode potentials

- they carry a great deal of useful information; values for chosen half-cell reductions are in Table 2.A
- $E^{\ominus} > 0$: the reduced form is more stable than hydrogen gas
- $E^{\ominus} < 0$: the reduced form is less stable than hydrogen gas

### Top and bottom of the series

- **fluorine** has the highest value: $F_2$ is the most eager to be reduced to $F^-$, so $F_2$ is the strongest oxidising agent and $F^-$ the weakest reducing agent
- at the bottom sits **lithium**: in water, $Li^+$ is the poorest oxidising agent and Li metal the strongest reducing agent

### Going down the table

- standard electrode potential falls
- oxidising power of the left-hand species falls; reducing power of the right-hand species rises

### Uses of electrochemical cells

- finding the pH of solutions, solubility products, equilibrium constants and other thermodynamic quantities
- potentiometric titrations

### Intext questions

**Q1** How could the standard electrode potential of the $Mg^{2+}|Mg$ system be found?

**Q2** Is it safe to keep copper sulphate solution in a zinc pot?

**Q3** Using the table of standard electrode potentials, name three substances able to oxidise ferrous ions under suitable conditions.

## 2.4 Nernst Equation

- so far every species in the electrode reaction was taken at unit concentration, which need not hold
- Nernst showed that for $M^{n+}(aq) + ne^- \rightarrow M(s)$, the electrode potential at any concentration, measured against the SHE, is:

$$E_{(M^{n+}/M)} = E^{\ominus}_{(M^{n+}/M)} - \frac{RT}{nF} \ln \frac{[M]}{[M^{n+}]}$$

- the concentration of solid M is taken as unity, giving:

$$E_{(M^{n+}/M)} = E^{\ominus}_{(M^{n+}/M)} - \frac{RT}{nF} \ln \frac{1}{[M^{n+}]}$$

- $E^{\ominus}_{(M^{n+}/M)}$ as defined above; $R$ = gas constant (8.314 J $K^{-1}$ $mol^{-1}$); $F$ = Faraday constant (96487 C $mol^{-1}$); $T$ = temperature in kelvin; $[M^{n+}]$ = concentration of $M^{n+}$

## TABLE 2.A Standard Electrode Potentials at 298 K

| Half-cell reaction (oxidised form + $ne^-$ → reduced form) | $E^\ominus$/V |
|---|---|
| $F_2(g) + 2e^- \rightarrow 2F^-$ | 2.87 |
| $Co^{3+} + e^- \rightarrow Co^{2+}$ | 1.81 |
| $H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$ | 1.78 |
| $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$ | 1.51 |
| $Au^{3+} + 3e^- \rightarrow Au(s)$ | 1.40 |
| $Cl_2(g) + 2e^- \rightarrow 2Cl^-$ | 1.36 |
| $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$ | 1.33 |
| $O_2(g) + 4H^+ + 4e^- \rightarrow 2H_2O$ | 1.23 |
| $MnO_2(s) + 4H^+ + 2e^- \rightarrow Mn^{2+} + 2H_2O$ | 1.23 |
| $Br_2 + 2e^- \rightarrow 2Br^-$ | 1.09 |
| $NO_3^- + 4H^+ + 3e^- \rightarrow NO(g) + 2H_2O$ | 0.97 |
| $2Hg^{2+} + 2e^- \rightarrow Hg_2^{2+}$ | 0.92 |
| $Ag^+ + e^- \rightarrow Ag(s)$ | 0.80 |
| $Fe^{3+} + e^- \rightarrow Fe^{2+}$ | 0.77 |
| $O_2(g) + 2H^+ + 2e^- \rightarrow H_2O_2$ | 0.68 |
| $I_2 + 2e^- \rightarrow 2I^-$ | 0.54 |
| $Cu^+ + e^- \rightarrow Cu(s)$ | 0.52 |
| $Cu^{2+} + 2e^- \rightarrow Cu(s)$ | 0.34 |
| $AgCl(s) + e^- \rightarrow Ag(s) + Cl^-$ | 0.22 |
| $AgBr(s) + e^- \rightarrow Ag(s) + Br^-$ | 0.10 |
| $2H^+ + 2e^- \rightarrow H_2(g)$ | 0.00 |
| $Pb^{2+} + 2e^- \rightarrow Pb(s)$ | −0.13 |
| $Sn^{2+} + 2e^- \rightarrow Sn(s)$ | −0.14 |
| $Ni^{2+} + 2e^- \rightarrow Ni(s)$ | −0.25 |
| $Fe^{2+} + 2e^- \rightarrow Fe(s)$ | −0.44 |
| $Cr^{3+} + 3e^- \rightarrow Cr(s)$ | −0.74 |
| $Zn^{2+} + 2e^- \rightarrow Zn(s)$ | −0.76 |
| $2H_2O + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$ | −0.83 |
| $Al^{3+} + 3e^- \rightarrow Al(s)$ | −1.66 |
| $Mg^{2+} + 2e^- \rightarrow Mg(s)$ | −2.36 |
| $Na^+ + e^- \rightarrow Na(s)$ | −2.71 |
| $Ca^{2+} + 2e^- \rightarrow Ca(s)$ | −2.87 |
| $K^+ + e^- \rightarrow K(s)$ | −2.93 |
| $Li^+ + e^- \rightarrow Li(s)$ | −3.05 |

### How to read the table

- ions are aqueous species and $H_2O$ is liquid; (g) marks gases and (s) marks solids
- reading down the table, $E^\ominus$ falls steadily from +2.87 V to −3.05 V
- towards the top, the left-hand (oxidised) species become stronger **oxidising agents**; towards the bottom, the right-hand (reduced) species become stronger **reducing agents**

### Sign of $E^\ominus$

- negative $E^\ominus$: the redox couple is a **stronger** reducing agent than the $H^+/H_2$ couple
- positive $E^\ominus$: the redox couple is a **weaker** reducing agent than the $H^+/H_2$ couple

## Nernst equation for the Daniell cell

- electrode potentials at any concentrations of $Cu^{2+}$ and $Zn^{2+}$:

cathode: $E_{(Cu^{2+}/Cu)} = E^{\ominus}_{(Cu^{2+}/Cu)} - \frac{RT}{2F} \ln \frac{1}{[Cu^{2+}(aq)]}$

anode: $E_{(Zn^{2+}/Zn)} = E^{\ominus}_{(Zn^{2+}/Zn)} - \frac{RT}{2F} \ln \frac{1}{[Zn^{2+}(aq)]}$

- cell potential = cathode potential minus anode potential:

$$E_{(cell)} = E^{\ominus}_{(Cu^{2+}/Cu)} - \frac{RT}{2F} \ln \frac{1}{[Cu^{2+}(aq)]} - E^{\ominus}_{(Zn^{2+}/Zn)} + \frac{RT}{2F} \ln \frac{1}{[Zn^{2+}(aq)]}$$

$$= E^{\ominus}_{(Cu^{2+}/Cu)} - E^{\ominus}_{(Zn^{2+}/Zn)} - \frac{RT}{2F} \left[ \ln \frac{1}{[Cu^{2+}(aq)]} - \ln \frac{1}{[Zn^{2+}(aq)]} \right]$$

$$\boxed{E_{(cell)} = E^{\ominus}_{(cell)} - \frac{RT}{2F} \ln \frac{[Zn^{2+}]}{[Cu^{2+}]}}$$

## What controls $E_{(cell)}$

- depends on the concentrations of both $Cu^{2+}$ and $Zn^{2+}$
- rises as $[Cu^{2+}]$ rises and as $[Zn^{2+}]$ falls

## At 298 K, in base-10 logs

- switching ln to log and putting in $R$, $F$ and $T$ = 298 K:

$$E_{(cell)} = E^{\ominus}_{(cell)} - \frac{0.059}{2} \log \frac{[Zn^{2+}]}{[Cu^{2+}]}$$

## Same $n$ for both electrodes

- the same number of electrons must be used for both electrodes; for the cell $Ni(s)|Ni^{2+}(aq)||Ag^{+}(aq)|Ag$
- cell reaction: $Ni(s) + 2Ag^{+}(aq) \rightarrow Ni^{2+}(aq) + 2Ag(s)$

$$E_{(cell)} = E^{\ominus}_{(cell)} - \frac{RT}{2F} \ln \frac{[Ni^{2+}]}{[Ag^{+}]^2}$$

## General form

- for a reaction $aA + bB \rightarrow cC + dD$ involving $n$ electrons:

$$\boxed{E_{(cell)} = E^{\ominus}_{(cell)} - \frac{RT}{nF} \ln Q}$$

$$= E^{\ominus}_{(cell)} - \frac{RT}{nF} \ln \frac{[C]^c\,[D]^d}{[A]^a\,[B]^b}$$

# 2.5 Equilibrium Constant from Nernst Equation

### A Daniell cell run down to equilibrium

- close the circuit (Figure 2.A) and $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$ proceeds
- over time $[Zn^{2+}]$ keeps climbing, $[Cu^{2+}]$ keeps dropping, and the voltmeter reading keeps falling
- eventually the concentrations stop changing and the voltmeter reads zero: **equilibrium** has been reached

### Nernst equation at equilibrium

$$E_{(cell)} = 0 = E^{\ominus}_{(cell)} - \frac{2.303RT}{2F} \log \frac{[Zn^{2+}]}{[Cu^{2+}]}$$

$$\text{or } E^{\ominus}_{(cell)} = \frac{2.303RT}{2F} \log \frac{[Zn^{2+}]}{[Cu^{2+}]}$$

- at equilibrium $[Zn^{2+}]/[Cu^{2+}] = K_c$ for this reaction

### $K_c$ for the Daniell reaction at 298 K

$$E^{\ominus}_{(cell)} = \frac{0.059}{2} \text{ V} \log K_C = 1.1 \text{ V}$$

$$\log K_C = \frac{(1.1 \text{ V} \times 2)}{0.059 \text{ V}} = 37.288 \Rightarrow \mathbf{K_C = 2 \times 10^{37}} \text{ at 298 K}$$

### In general

$$\boxed{E^{\ominus}_{(cell)} = \frac{2.303RT}{nF} \log K_C}$$

- links a reaction's equilibrium constant to the standard potential of the cell in which it runs
- equilibrium constants that are hard to measure directly can be calculated from the cell's $E^{\ominus}$

### Worked example emf at non-standard concentrations

For the reaction $Mg(s) + 2Ag^{+}(0.0001M) \rightarrow Mg^{2+}(0.130M) + 2Ag(s)$, write the cell and find $E_{(cell)}$, given $E^{\ominus}_{(cell)} = 3.17$ V.

- cell: $Mg|Mg^{2+}(0.130M)||Ag^{+}(0.0001M)|Ag$

$$E_{(cell)} = E^{\ominus}_{(cell)} - \frac{RT}{2F} \ln \frac{[Mg^{2+}]}{[Ag^{+}]^2} = 3.17 \text{ V} - \frac{0.059 \text{ V}}{2} \log \frac{0.130}{(0.0001)^2} = 3.17 \text{ V} - 0.21 \text{ V} = \mathbf{2.96 \text{ V}}$$

### *Worked example* equilibrium constant from $E^{\ominus}$

Find the equilibrium constant of $Cu(s) + 2Ag^+(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$, for which $E^{\ominus}_{(cell)} = 0.46$ V.

$$E^{\ominus}_{(cell)} = \frac{0.059}{2}\ \text{V}\ \log K_C = 0.46\ \text{V} \Rightarrow \log K_C = \frac{0.46\ \text{V} \times 2}{0.059\ \text{V}} = 15.6 \Rightarrow \mathbf{K_C = 3.92 \times 10^{15}}$$

## 2.6 Electrochemical Cell and Gibbs Energy of the Reaction

### *Work from a cell*

- electrical work per second = electrical potential × total charge passed
- to get the **maximum** work from a galvanic cell, charge must pass reversibly
- the reversible work a galvanic cell does equals the fall in its Gibbs energy
- with emf $E$, charge passed $nF$ and Gibbs energy of reaction $\Delta_r G$:

$$\Delta_r G = -nFE_{(cell)}$$

### *Intensive vs extensive*

- $E_{(cell)}$ is intensive; $\Delta_r G$ is extensive and its value depends on $n$
- $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$: $\Delta_r G = -2FE_{(cell)}$
- $2Zn(s) + 2Cu^{2+}(aq) \rightarrow 2Zn^{2+}(aq) + 2Cu(s)$: $\Delta_r G = -4FE_{(cell)}$

### *Standard Gibbs energy from $E^{\ominus}$*

- with every reacting species at unit concentration, $E_{(cell)} = E^{\ominus}_{(cell)}$, so:

$$\Delta_r G^{\ominus} = -nFE^{\ominus}_{(cell)}$$

- measuring $E^{\ominus}_{(cell)}$ therefore yields an important thermodynamic quantity, the standard Gibbs energy of reaction $\Delta_r G^{\ominus}$
- and from that the equilibrium constant follows: $\Delta_r G^{\ominus} = -RT \ln K$

### *Worked example* standard Gibbs energy of the Daniell reaction

The Daniell cell has a standard electrode potential of 1.1 V. Find the standard Gibbs energy for $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$.

- $\Delta_r G^{\ominus} = -nFE^{\ominus}_{(cell)}$, with $n = 2$, $F = 96487\ \text{C mol}^{-1}$, $E^{\ominus}_{(cell)} = 1.1$ V

$$\Delta_r G^{\ominus} = -2 \times 1.1\ \text{V} \times 96487\ \text{C mol}^{-1} = -212271\ \text{J mol}^{-1} = \mathbf{-212.27\ kJ\ mol^{-1}}$$

### Intext questions

**Q4** A hydrogen electrode dips into a solution of pH 10. What is its potential?

**Q5** Find the emf of the cell in which Ni(s) + 2Ag$^+$(0.002 M) → Ni$^{2+}$(0.160 M) + 2Ag(s) occurs, given $E^\ominus_{cell}$ = 1.05 V.

**Q6** A cell running the reaction 2Fe$^{3+}$(aq) + 2I$^-$(aq) → 2Fe$^{2+}$(aq) + I$_2$(s) has $E^\ominus_{cell}$ = 0.236 V at 298 K. Work out $\Delta_r G^\ominus$ and $K$ for this reaction.

## 2.7 Conductance of Electrolytic Solutions

A few terms need defining before looking at how electrolytic solutions conduct electricity.

### Resistance, R

- measured in ohm (Ω); in SI base units, Ω = (kg m$^2$)/(s$^3$ A$^2$)
- can be measured with a Wheatstone bridge, familiar from physics
- directly proportional to length $l$, inversely proportional to cross-sectional area $A$:

$$R \propto \frac{l}{A} \quad \text{or} \quad \boxed{R = \rho \frac{l}{A}}$$

### Resistivity, ϱ

- the constant ρ (Greek rho) is **resistivity** (specific resistance); IUPAC prefers "resistivity", used from here on
- SI unit ohm metre (Ω m); the submultiple ohm centimetre (Ω cm) is also common
- physically: the resistance of a piece 1 m long with 1 m$^2$ cross-section
- 1 Ω m = 100 Ω cm, or 1 Ω cm = 0.01 Ω m

### Conductance, G

- the reciprocal of resistance:

$$\boxed{G = \frac{1}{R} = \frac{A}{\rho\, l} = \kappa \frac{A}{l}}$$

- SI unit **siemens** (S), equal to ohm$^{-1}$ (also called mho) or Ω$^{-1}$

### Conductivity, κ

- the reciprocal of resistivity: **conductivity** (specific conductance), symbol κ (Greek kappa); IUPAC prefers "conductivity"
- SI unit S m$^{-1}$, though κ is often given in S cm$^{-1}$; 1 S cm$^{-1}$ = 100 S m$^{-1}$
- in S m$^{-1}$, the conductance of a piece 1 m long with 1 m$^2$ cross-section

Table 2.B The values of Conductivity of some Selected Materials at 298.15 K

| Material | Conductivity / S $m^{-1}$ | Material | Conductivity / S $m^{-1}$ |
|---|---|---|---|
| *Conductors* | | *Aqueous solutions* | |
| Sodium | $2.1 \times 10^3$ | Pure water | $3.5 \times 10^{-5}$ |
| Copper | $5.9 \times 10^3$ | 0.1 M HCl | 3.91 |
| Silver | $6.2 \times 10^3$ | 0.01 M KCl | 0.14 |
| Gold | $4.5 \times 10^3$ | 0.01 M NaCl | 0.12 |
| Iron | $1.0 \times 10^3$ | 0.1 M HAc | 0.047 |
| Graphite | $1.2 \times 10$ | 0.01 M HAc | 0.016 |
| *Insulators* | | *Semiconductors* | |
| Glass | $1.0 \times 10^{-16}$ | CuO | $1 \times 10^{-7}$ |
| Teflon | $1.0 \times 10^{-18}$ | Si | $1.5 \times 10^{-2}$ |
| | | Ge | 2.0 |

### Conductivity varies enormously

- as Table 2.B shows, it spans a huge range and depends on the material, and also on the temperature and pressure of measurement
- its size sorts materials into **conductors**, **insulators** and **semiconductors**

### The classes

- **conductors**: metals and alloys, with very high conductivity; some non-metals (carbon-black, graphite, certain organic polymers) also conduct electronically
- **insulators**: very low conductivity, e.g. glass, ceramics
- **semiconductors**: in between, e.g. silicon, doped silicon, gallium arsenide; key electronic materials

### Superconductors

- by definition have zero resistivity, or infinite conductivity
- once known only among metals and alloys at very low temperatures (0 to 15 K)
- now many ceramics and mixed oxides superconduct at up to 150 K

### Metallic (electronic) conductance

- conduction through metals, carried by moving electrons; it depends on:
- (i) what the metal is and how it is structured; (ii) how many valence electrons each atom has
- (iii) temperature: it falls as temperature rises

### Electronically conducting polymers

- 1977: MacDiarmid, Heeger and Shirakawa found that acetylene gas polymerises to polyacetylene, which gains metallic lustre and conductivity on exposure to iodine vapour; polyaniline, polypyrrole and polythiophene followed
- built wholly from elements such as carbon and hydrogen, sometimes with nitrogen, oxygen or sulphur, these metal-like polymers are far lighter than ordinary metals and suit light-weight batteries
- they keep polymer mechanical properties such as flexibility, so devices like transistors can bend like a plastic sheet; the three discoverers shared the 2000 Nobel Prize in Chemistry

### Metals and semiconductors

- electrons enter a metallic conductor at one end and leave at the other, so its composition does not change
- conduction in semiconductors works by a more complex mechanism

### Electrolytic (ionic) conductance

- even very pure water holds a little $H^+$ and $OH^-$ (~$10^{-7}$ M), giving it a very low conductivity ($3.5 \times 10^{-5}$ S $m^{-1}$)
- dissolved electrolytes add their own ions, raising the conductivity
- conduction by ions in solution is **electrolytic** or **ionic** conductance

### What ionic conductivity depends on

- (i) the kind of electrolyte added
- (ii) the size of its ions and how they are solvated
- (iii) what the solvent is, and how viscous
- (iv) the concentration of electrolyte
- (v) temperature: it **rises** as temperature rises

Passing direct current through an ionic solution for a long time can change its composition, because electrochemical reactions take place.

## 2.8 Measurement of the Conductivity of Ionic Solutions

### Two problems, two fixes

- a Wheatstone bridge measures an unknown resistance accurately, but an ionic solution raises two difficulties
- (1) DC changes the solution's composition: solved by using an **alternating current (AC)** source
- (2) a solution cannot be wired into the bridge like a metal wire or other solid: solved by a purpose-built vessel, the **conductivity cell**, made in several designs (two simple ones in Figure 2.D)

**Figure 2.D** Two different types of conductivity cells.

- each is a glass vessel holding the solution, with two platinized Pt electrodes facing each other inside
- connecting wires lead from the electrodes up and out of the vessel to the measuring circuit

### Inside the conductivity cell

- two platinum electrodes coated with platinum black (finely divided Pt deposited electrochemically on them)
- each has cross-sectional area $A$; they are a distance $l$ apart
- the solution between them is thus a column of length $l$ and cross-section $A$, whose resistance is:

$$R = \rho \frac{l}{A} = \frac{l}{\kappa A}$$

## Cell constant, $G^*$

- $l/A$ is the **cell constant**, $G^*$; it depends on electrode spacing and area and has dimension length$^{-1}$
- it could be worked out from $l$ and $A$, but measuring these is awkward and unreliable
- instead, measure the cell's resistance when filled with a solution of known conductivity, usually KCl, whose conductivity is accurately known at many concentrations (Table 2.C) and temperatures:

$$G^* = \frac{l}{A} = R\kappa$$

- once known, $G^*$ serves for measuring the resistance or conductivity of any solution

TABLE 2.C **Conductivity and Molar conductivity of KCl solutions at 298.15K**

| Concentration (molarity) | | Conductivity | | Molar conductivity | |
|---|---|---|---|---|---|
| mol $L^{-1}$ | mol $m^{-3}$ | S $cm^{-1}$ | S $m^{-1}$ | S $cm^2$ $mol^{-1}$ | S $m^2$ $mol^{-1}$ |
| 1.000 | 1000 | 0.1113 | 11.13 | 111.3 | 111.3 $\times 10^{-4}$ |
| 0.100 | 100.0 | 0.0129 | 1.29 | 129.0 | 129.0 $\times 10^{-4}$ |
| 0.010 | 10.00 | 0.00141 | 0.141 | 141.0 | 141.0 $\times 10^{-4}$ |

## The measuring bridge

- two fixed resistances $R_3$ and $R_4$, a variable resistance $R_1$, and the conductivity cell of unknown resistance $R_2$ (Figure 2.E)
- powered by an oscillator O, an a.c. source at audio frequencies of 550 to 5000 cycles per second
- P is a detector (a headphone or other electronic device); the bridge is balanced when no current passes through it, and then:

$$\text{unknown resistance } R_2 = \frac{R_1 R_4}{R_3}$$

**Figure 2.E** Arrangement for measurement of resistance of a solution of an electrolyte.

- a diamond-shaped Wheatstone bridge: the conductivity cell ($R_2$) and variable resistance $R_1$ on the upper arms, $R_4$ and $R_3$ on the lower arms
- the detector P bridges the middle; the oscillator O feeds the circuit at the bottom

## From resistance to conductivity

- cheap conductivity meters now read the conductance or resistance of the solution in the cell directly
- knowing the cell constant and the solution's resistance:

$$\kappa = \frac{\text{cell constant}}{R} = \frac{G^*}{R}$$

In one solvent at one temperature, solutions of different electrolytes differ in conductivity because of the charge and size of the ions they form.

### Why a new quantity is needed

- conductivity also depends on how many ions the electrolyte forms, their concentration, and how easily they move under a potential gradient
- a more physically meaningful measure is therefore defined: **molar conductivity**, $\Lambda_m$ (Greek lambda)

### Molar conductivity, $\Lambda_m$

$$\text{Molar conductivity} = \Lambda_m = \frac{\kappa}{c}$$

- $\kappa$ in S m$^{-1}$ and $c$ in mol m$^{-3}$ give $\Lambda_m$ in S m$^2$ mol$^{-1}$
- since 1 mol m$^{-3}$ = 1000 (L/m$^3$) × molarity (mol/L):

$$\Lambda_m(\text{S m}^2\text{ mol}^{-1}) = \frac{\kappa\ (\text{S m}^{-1})}{1000\ \text{L m}^{-3} \times \text{molarity (mol L}^{-1})}$$

- $\kappa$ in S cm$^{-1}$ and concentration in mol cm$^{-3}$ give $\Lambda_m$ in S cm$^2$ mol$^{-1}$:

$$\Lambda_m(\text{S cm}^2\text{ mol}^{-1}) = \frac{\kappa\ (\text{S cm}^{-1}) \times 1000\ (\text{cm}^3/\text{L})}{\text{molarity (mol/L)}}$$

- both units appear in the literature: 1 S m$^2$ mol$^{-1}$ = 10$^4$ S cm$^2$ mol$^{-1}$, or 1 S cm$^2$ mol$^{-1}$ = 10$^{-4}$ S m$^2$ mol$^{-1}$

### Worked example cell constant, $\kappa$ and $\Lambda_m$

A conductivity cell filled with 0.1 mol L$^{-1}$ KCl (conductivity 1.29 S/m) has a resistance of 100 **Ω**. Filled with 0.02 mol L$^{-1}$ KCl, its resistance is 520 **Ω**. Find the conductivity and molar conductivity of the 0.02 mol L$^{-1}$ solution.

- cell constant $G^*$ = conductivity × resistance = 1.29 S/m × 100 Ω = 129 m$^{-1}$ = 1.29 cm$^{-1}$
- $\kappa$(0.02 mol L$^{-1}$) = $G^*/R$ = 129 m$^{-1}$ / 520 Ω = **0.248 S m$^{-1}$**
- concentration = 0.02 mol L$^{-1}$ = 1000 × 0.02 mol m$^{-3}$ = 20 mol m$^{-3}$

$$\Lambda_m = \frac{\kappa}{c} = \frac{248 \times 10^{-3}\ \text{S m}^{-1}}{20\ \text{mol m}^{-3}} = \mathbf{124 \times 10^{-4}\ S\ m^2\ mol^{-1}}$$

**Alternatively, in cm units**

$$\kappa = \frac{1.29\ \text{cm}^{-1}}{520\ \Omega} = 0.248 \times 10^{-2}\ \text{S cm}^{-1}$$

$$\Lambda_m = \kappa \times 1000\ \text{cm}^3\ \text{L}^{-1}\ \text{molarity}^{-1}$$

$$= \frac{0.248 \times 10^{-2}\ \text{S cm}^{-1} \times 1000\ \text{cm}^3\ \text{L}^{-1}}{0.02\ \text{mol L}^{-1}} = \mathbf{124\ S\ cm^2\ mol^{-1}}$$

### *Worked example* resistivity, conductivity and $\Lambda_m$ of NaOH

A column of 0.05 mol $L^{-1}$ NaOH solution, 1 cm in diameter and 50 cm long, has a resistance of 5.55 × $10^3$ ohm. Find its resistivity, conductivity and molar conductivity.

**In cm units**

- $A = \pi r^2 = 3.14 \times 0.5^2\ cm^2 = 0.785\ cm^2 = 0.785 \times 10^{-4}\ m^2$; $l = 50\ cm = 0.5\ m$

$$R = \frac{\rho l}{A} \Rightarrow \rho = \frac{RA}{l} = \frac{5.55 \times 10^3\ \Omega \times 0.785\ cm^2}{50\ cm} = \mathbf{87.135\ \Omega\ cm}$$

$$\kappa = \frac{1}{\rho} = \frac{1}{87.135}\ S\ cm^{-1} = \mathbf{0.01148\ S\ cm^{-1}}$$

$$\Lambda_m = \frac{\kappa \times 1000\ cm^3\ L^{-1}}{c} = \frac{0.01148\ S\ cm^{-1} \times 1000\ cm^3\ L^{-1}}{0.05\ mol\ L^{-1}} = \mathbf{229.6\ S\ cm^2\ mol^{-1}}$$

**The same in m units**

$$\rho = \frac{RA}{l} = \frac{5.55 \times 10^3\ \Omega \times 0.785 \times 10^{-4}\ m^2}{0.5\ m} = 87.135 \times 10^{-2}\ \Omega\ m$$

$$\kappa = \frac{1}{\rho} = \frac{100}{87.135}\ \Omega^{-1}\ m^{-1} = 1.148\ S\ m^{-1}$$

$$\Lambda_m = \frac{\kappa}{c} = \frac{1.148\ S\ m^{-1}}{50\ mol\ m^{-3}} = 229.6 \times 10^{-4}\ S\ m^2\ mol^{-1}$$

## 2.9 Variation of Conductivity and Molar Conductivity with Concentration

### *Conductivity falls on dilution*

- both κ and $\Lambda_m$ change with electrolyte concentration
- κ always **decreases** as concentration decreases, for weak and strong electrolytes alike
- reason: dilution leaves fewer current-carrying ions in each unit volume
- at a given concentration, κ is the conductance of one unit volume of solution held between two platinum electrodes of unit cross-sectional area placed unit length apart

### $\kappa$ as a conductance

$$G = \frac{\kappa A}{l} = \kappa$$

- since $A$ and $l$ are both unity, in m or cm as appropriate

### $\Lambda_m$ as a conductance

- the conductance of the volume $V$ of solution holding one mole of electrolyte, between electrodes of area $A$ at unit distance

$\Lambda_m = \frac{\kappa A}{l}$; with $l = 1$ and $A = V$: $\boxed{\Lambda_m = \kappa V}$

### Molar conductivity rises on dilution

- $\Lambda_m$ **increases** as concentration decreases, because the volume $V$ containing one mole of electrolyte grows
- the drop in $\kappa$ on dilution is more than made up for by the increase in volume
- physically: $\Lambda_m$ is the conductance of the solution between electrodes at unit distance whose area is big enough to hold the volume containing one mole of electrolyte

### Limiting molar conductivity

- as concentration approaches zero, $\Lambda_m$ becomes the **limiting molar conductivity**, $\Lambda_m^\circ$
- strong and weak electrolytes show different variation of $\Lambda_m$ with concentration (Figure 2.F)

### Strong electrolytes

- $\Lambda_m$ climbs slowly on dilution, following:

$$\boxed{\Lambda_m = \Lambda_m^\circ - A\,c^{1/2}}$$

- a plot of $\Lambda_m$ against $c^{1/2}$ is a straight line: intercept $\Lambda_m^\circ$, slope '$-A$'
- for a given solvent and temperature, $A$ depends on the electrolyte type, i.e. the charges on the cation and anion it forms
- NaCl, $CaCl_2$, $MgSO_4$ are 1-1, 2-1 and 2-2 electrolytes; every electrolyte of one type has the same $A$

**Figure 2.F** Molar conductivity versus $c^{1/2}$ for acetic acid (weak electrolyte) and potassium chloride (strong electrolyte) in aqueous solutions.

- $\Lambda_m$ (S cm$^2$ mol$^{-1}$) on the vertical axis, $c^{1/2}$ on the horizontal
- **KCl**: high values on a gently falling straight line that can be extended back to meet the axis at $\Lambda_m^\circ$
- **acetic acid**: low values over most of the range, climbing very steeply only as $c^{1/2}$ nears zero

***Worked example*** $\Lambda_m^\circ$ and $A$ for KCl

The molar conductivities of KCl solutions at 298 K are tabulated below. Show that $\Lambda_m$ plotted against $c^{1/2}$ gives a straight line, and find $\Lambda_m^\circ$ and $A$ for KCl.

| $c$ / mol L$^{-1}$ | $\Lambda_m$ / S cm$^2$ mol$^{-1}$ |
|---|---|
| 0.000198 | 148.61 |
| 0.000309 | 148.29 |
| 0.000521 | 147.81 |
| 0.000989 | 147.09 |

- take the square root of each concentration:

| $c^{1/2}$ / (mol L$^{-1}$)$^{1/2}$ | $\Lambda_m$ / S cm$^2$ mol$^{-1}$ |
|---|---|
| 0.01407 | 148.61 |
| 0.01758 | 148.29 |
| 0.02283 | 147.81 |
| 0.03145 | 147.09 |

- $\Lambda_m$ (y-axis) against $c^{1/2}$ (x-axis), plotted in Figure 2.G, is very nearly a straight line
- intercept at $c^{1/2} = 0$: **$\Lambda_m^\circ$ = 150.0 S cm$^2$ mol$^{-1}$**
- **$A$ = – slope = 87.46 S cm$^2$ mol$^{-1}$/(mol L$^{-1}$)$^{1/2}$**

**Figure 2.G** Variation of $\Lambda_m$ against $c^{½}$.

- the four KCl points, $\Lambda_m$ (about 147 to 149 S cm$^2$ mol$^{-1}$) against $c^{½}$ (about 0.014 to 0.031), fall on a straight line sloping gently downward
- the line, extended back to $c^{½} = 0$, meets the vertical axis at 150.0 S cm$^2$ mol$^{-1}$ ($\Lambda_m^\circ$); the negative of its slope gives $A$

## Kohlrausch's regularities

- studying $\Lambda_m^\circ$ for many strong electrolytes, Kohlrausch noticed that $\Lambda_m^\circ(KX) - \Lambda_m^\circ(NaX)$ is almost the same whatever X is; at 298 K:

$$\Lambda_m^\circ(KCl) - \Lambda_m^\circ(NaCl) = \Lambda_m^\circ(KBr) - \Lambda_m^\circ(NaBr) = \Lambda_m^\circ(KI) - \Lambda_m^\circ(NaI) \approx 23.4 \text{ S cm}^2 \text{ mol}^{-1}$$

$$\Lambda_m^\circ(NaBr) - \Lambda_m^\circ(NaCl) = \Lambda_m^\circ(KBr) - \Lambda_m^\circ(KCl) \approx 1.8 \text{ S cm}^2 \text{ mol}^{-1}$$

## Law of independent migration of ions

- **Kohlrausch law**: the limiting molar conductivity of an electrolyte is the sum of separate contributions from its anion and its cation
- with $\lambda^\circ_{Na^+}$ and $\lambda^\circ_{Cl^-}$ the limiting molar conductivities of sodium and chloride ions:

$$\Lambda_m^\circ(NaCl) = \lambda^\circ_{Na^+} + \lambda^\circ_{Cl^-}$$

- for an electrolyte giving $\nu_+$ cations and $\nu_-$ anions:

$$\Lambda_m^\circ = \nu_+ \lambda^\circ_+ + \nu_- \lambda^\circ_-$$

- $\lambda^\circ_+$, $\lambda^\circ_-$: limiting molar conductivities of cation and anion; values at 298 K in Table 2.D

TABLE 2.D Limiting Molar Conductivity for some Ions in Water at 298 K

| Ion | $\lambda^0$ / (S cm$^2$ mol$^{-1}$) | Ion | $\lambda^0$ / (S cm$^2$ mol$^{-1}$) |
|---|---|---|---|
| $H^+$ | 349.6 | $OH^-$ | 199.1 |
| $Na^+$ | 50.1 | $Cl^-$ | 76.3 |
| $K^+$ | 73.5 | $Br^-$ | 78.1 |
| $Ca^{2+}$ | 119.0 | $CH_3COO^-$ | 40.9 |
| $Mg^{2+}$ | 106.0 | $SO_4^{2-}$ | 160.0 |

## Weak electrolytes

- weak electrolytes such as acetic acid are only slightly dissociated at higher concentrations
- so on dilution $\Lambda_m$ changes because the degree of dissociation rises, and with it the number of ions in the volume holding 1 mol of electrolyte
- $\Lambda_m$ therefore rises **steeply** on dilution, most of all at low concentrations (Figure 2.F), and $\Lambda_m^\circ$ cannot be found by extrapolating to zero concentration
- at infinite dilution ($c \to$ zero) dissociation is complete ($\alpha = 1$), but the conductivity is then too small to measure accurately; $\Lambda_m^\circ$ for weak electrolytes comes instead from Kohlrausch law (see the worked example for HAc)

## Degree of dissociation and $K_a$

- at concentration $c$, the degree of dissociation $\alpha$ is approximately:

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ}$$

- for a weak electrolyte such as acetic acid (see Class XI, Unit 7):

$$K_a = \frac{c\alpha^2}{(1-\alpha)} = \frac{c\,\Lambda_m^{\,2}}{\Lambda_m^{\circ 2}(1-\Lambda_m/\Lambda_m^\circ)} = \frac{c\,\Lambda_m^{\,2}}{\Lambda_m^\circ(\Lambda_m^\circ - \Lambda_m)}$$

## Applications of Kohlrausch law

- $\Lambda_m^\circ$ of any electrolyte can be calculated from the $\lambda^\circ$ of its individual ions
- for weak electrolytes like acetic acid, the dissociation constant follows once $\Lambda_m^\circ$ and $\Lambda_m$ at a given concentration $c$ are known

### Worked example $\Lambda_m^\circ$ from ionic values

Use the ionic data in Table 2.D to find $\boldsymbol{\Lambda_m^\circ}$ for $CaCl_2$ and for $MgSO_4$.

$\Lambda_m^\circ(CaCl_2) = \lambda^\circ_{Ca^{2+}} + 2\lambda^\circ_{Cl^-} = 119.0 + 2(76.3) = (119.0 + 152.6) =$ **271.6 S cm$^2$ mol$^{-1}$**

$\Lambda_m^\circ(MgSO_4) = \lambda^\circ_{Mg^{2+}} + \lambda^\circ_{SO_4^{2-}} = 106.0 + 160.0 =$ **266 S cm$^2$ mol$^{-1}$**

### Worked example $\Lambda_m^\circ$ of a weak acid by combining strong electrolytes

$\boldsymbol{\Lambda_m^\circ}$ for NaCl, HCl and NaAc is 126.4, 425.9 and 91.0 S cm$^2$ mol$^{-1}$. Find $\boldsymbol{\Lambda_m^\circ}$ for HAc.

$$\Lambda_m^\circ(HAc) = \lambda^\circ_{H^+} + \lambda^\circ_{Ac^-} = \lambda^\circ_{H^+} + \lambda^\circ_{Cl^-} + \lambda^\circ_{Ac^-} + \lambda^\circ_{Na^+} - \lambda^\circ_{Cl^-} - \lambda^\circ_{Na^+}$$

$$= \Lambda_m^\circ(HCl) + \Lambda_m^\circ(NaAc) - \Lambda_m^\circ(NaCl) = (425.9 + 91.0 - 126.4) = \textbf{390.5 S cm}^2\textbf{ mol}^{-1}$$

### Worked example dissociation constant of acetic acid

0.001028 mol L$^{-1}$ acetic acid has a conductivity of $4.95 \times 10^{-5}$ S cm$^{-1}$. With $\boldsymbol{\Lambda_m^\circ}$ = 390.5 S cm$^2$ mol$^{-1}$, find its dissociation constant.

$$\Lambda_m = \frac{\kappa}{c} = \frac{4.95 \times 10^{-5}\ \text{S cm}^{-1} \times 1000\ \text{cm}^3\ \text{L}^{-1}}{0.001028\ \text{mol L}^{-1}} = 48.15\ \text{S cm}^2\ \text{mol}^{-1}$$

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} = \frac{48.15\ \text{S cm}^2\ \text{mol}^{-1}}{390.5\ \text{S cm}^2\ \text{mol}^{-1}} = 0.1233$$

$$K_a = \frac{c\alpha^2}{(1-\alpha)} = \frac{0.001028\ \text{mol L}^{-1} \times (0.1233)^2}{1 - 0.1233} = \textbf{1.78} \times \textbf{10}^{-5}\textbf{ mol L}^{-1}$$

### Intext questions

**Q7** Why is a solution's conductivity lowered when it is diluted?

**Q8** Propose a method for finding $\Lambda_m^\circ$ of water.

**Q9** 0.025 mol $L^{-1}$ methanoic acid has a molar conductivity of 46.1 S $cm^2$ $mol^{-1}$. Given $\lambda^0(H^+)$ = 349.6 S $cm^2$ $mol^{-1}$ and $\lambda^0(HCOO^-)$ = 54.6 S $cm^2$ $mol^{-1}$, find its degree of dissociation and dissociation constant.

## 2.10 Electrolytic Cells and Electrolysis

### What an electrolytic cell does

- an external voltage source drives a chemical reaction
- such processes matter greatly in the laboratory and in chemical industry
- one of the simplest: two copper strips in aqueous copper sulphate, with a DC voltage across them

### The copper cell

- at the cathode (negative), $Cu^{2+}$ ions are discharged and copper metal deposits:

$$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$$

- at the anode, copper passes into solution as $Cu^{2+}$:

$$Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$$

- copper is dissolved (oxidised) at the anode and deposited (reduced) at the cathode

### Refining and extracting metals

- copper purification: impure copper is the anode and dissolves as current flows; pure copper builds up on the cathode
- Na, Mg, Al and others are made on a large scale by electrochemical reduction of their cations, where no suitable chemical reducing agent exists
- Na and Mg: electrolysis of their fused chlorides; Al: electrolysis of aluminium oxide with cryolite present

### Faraday's laws of electrolysis

- Michael Faraday first described electrolysis quantitatively; his laws also follow from the ideas above. After wide study of solutions and melts, he published them in 1833–34
- **First law**: how much chemical change occurs at an electrode during electrolysis is proportional to the charge sent through the electrolyte, whether a solution or a melt
- **Second law**: when one and the same charge passes, the masses of different substances set free stand in the ratio of their chemical equivalent weights (metal's atomic mass divided by the electrons its cation needs for reduction)

### Measuring the charge

- Faraday had no constant-current sources; a **coulometer** (a standard electrolytic cell) was used, inferring the charge from the mass of metal, usually silver or copper, deposited or used up
- coulometers are now obsolete; with a constant current $I$, the charge passed is:

$$Q = It$$ ($Q$ in coulombs, $I$ in amperes, $t$ in seconds)

### Charge and stoichiometry

- the charge needed depends on the stoichiometry of the electrode reaction
- $Ag^+(aq) + e^- \rightarrow Ag(s)$: one mole of electrons reduces one mole of silver ions
- $Mg^{2+}(l) + 2e^- \rightarrow Mg(s)$ needs 2 mol of electrons ($2F$); $Al^{3+}(l) + 3e^- \rightarrow Al(s)$ needs 3 mol ($3F$)

### The Faraday, $F$

- the charge on one electron is $1.6021 \times 10^{-19}$ C, so one mole of electrons carries:

$$N_A \times 1.6021 \times 10^{-19}\ \text{C} = 6.02 \times 10^{23}\ \text{mol}^{-1} \times 1.6021 \times 10^{-19}\ \text{C} = \mathbf{96487\ C\ mol^{-1}}$$

- this quantity of electricity is the **Faraday**, symbol $F$; for rough work, $1F \approx 96500\ \text{C mol}^{-1}$
- charge passed = current (A) × time (s)
- commercial metal production uses currents as high as 50,000 amperes, about 0.518 $F$ per second

### Worked example mass of copper deposited

**$CuSO_4$ solution is electrolysed with 1.5 amperes for 10 minutes. What mass of copper is deposited on the cathode?**

- $t = 600$ s; charge = current × time = 1.5 A × 600 s = 900 C
- $Cu^{2+}(aq) + 2e^- = Cu(s)$: $2F$, i.e. $2 \times 96487$ C, deposits 1 mol (63 g) of Cu

mass for 900 C = $(63\ \text{g mol}^{-1} \times 900\ \text{C}) / (2 \times 96487\ \text{C mol}^{-1})$ = **0.2938 g**

## 2.11 Products of Electrolysis

- what forms depends on the material being electrolysed and on the kind of electrodes
- an **inert** electrode (platinum, gold) takes no part in the reaction and only supplies or accepts electrons; a **reactive** electrode takes part in the electrode reaction
- so reactive and inert electrodes may give different products

## What decides the products

- the oxidising and reducing species present in the cell, and their standard electrode potentials
- kinetics: some feasible processes are so slow that at lower voltages they seem not to happen; an extra potential, the **overpotential**, must be applied, making them harder to bring about

## Molten vs aqueous NaCl

- **molten NaCl**: sodium metal and $Cl_2$ gas; the only cation, $Na^+$, is reduced at the cathode ($Na^+ + e^- \rightarrow Na$) and the only anion, $Cl^-$, is oxidised at the anode ($Cl^- \rightarrow \frac{1}{2}Cl_2 + e^-$)
- **aqueous NaCl**: NaOH, $Cl_2$ and $H_2$; besides $Na^+$ and $Cl^-$, $H^+$, $OH^-$ and solvent $H_2O$ are present

## At the cathode

- two reductions compete:

$Na^+(aq) + e^- \rightarrow Na(s) \quad E^{\ominus}_{(cell)} = -2.71\ V$

$H^+(aq) + e^- \rightarrow \frac{1}{2}H_2(g) \quad E^{\ominus}_{(cell)} = 0.00\ V$

- the one with the **higher** $E^{\ominus}$ wins, so $H^+$ is reduced; $H^+$ comes from $H_2O(l) \rightarrow H^+(aq) + OH^-(aq)$
- adding the two gives the net cathode reaction:

$$H_2O(l) + e^- \rightarrow \tfrac{1}{2}H_2(g) + OH^-$$

## At the anode

- two oxidations are possible:

$Cl^-(aq) \rightarrow \frac{1}{2}Cl_2(g) + e^- \quad E^{\ominus}_{(cell)} = 1.36\ V$

$2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^- \quad E^{\ominus}_{(cell)} = 1.23\ V$

- the one with the **lower** $E^{\ominus}$ should win, so water ought to be oxidised before $Cl^-$
- but the **overpotential of oxygen** makes chloride oxidation the preferred reaction

## Aqueous NaCl: overall

$NaCl(aq) \rightarrow Na^+(aq) + Cl^-(aq)$ (in $H_2O$)
cathode: $H_2O(l) + e^- \rightarrow \frac{1}{2}H_2(g) + OH^-(aq)$
anode: $Cl^-(aq) \rightarrow \frac{1}{2}Cl_2(g) + e^-$

net: $NaCl(aq) + H_2O(l) \rightarrow Na^+(aq) + OH^-(aq) + \frac{1}{2}H_2(g) + \frac{1}{2}Cl_2(g)$

- to allow for concentration, the Nernst-equation electrode potentials replace the standard ones

## Electrolysis of sulphuric acid

- possible anode processes include:

$$2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^- \quad E^{\ominus}_{(cell)} = +1.23\ V$$

### Sulphuric acid: the competing anode reactions

$$2SO_4^{2-}(aq) \rightarrow S_2O_8^{2-}(aq) + 2e^- \quad E^{\ominus}_{(cell)} = 1.96\ V$$

- in **dilute** sulphuric acid, oxidation of water to $O_2$ (+1.23 V) is preferred
- at **higher** $H_2SO_4$ concentrations, oxidation of sulphate to $S_2O_8^{2-}$ is preferred

### Intext questions

**Q10** A metal wire carries 0.5 ampere for 2 hours. How many electrons pass through it?

**Q11** List some metals that are obtained by electrolysis.

**Q12** For $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$, how many coulombs are needed to reduce 1 mol of $Cr_2O_7^{2-}$?

## 2.12 Batteries

- any battery (really one cell, or several in series) or cell used as a source of electricity is at heart a **galvanic cell**: chemical energy of a redox reaction becomes electrical energy
- to be practical it must be fairly light and compact, and its voltage must stay nearly steady in use; there are two main types

## 2.13 Primary Batteries

### Primary: use once

- the reaction happens only once; after a period of use the battery is dead and cannot be reused
- best-known example: the **dry cell** (the **Leclanché cell**, after its discoverer), found in transistors and clocks

**Figure 2.H** A commercial dry cell consists of a graphite (carbon) cathode in a zinc container; the latter acts as the anode.

- a zinc cup (anode) forms the outer case; a carbon rod (cathode) runs down the centre
- around the rod: $MnO_2$ + carbon black, with $NH_4Cl$ paste filling the space

### Dry cell

- zinc container = anode; carbon (graphite) rod in powdered $MnO_2$ and carbon = cathode (Figure 2.H); a moist paste of $NH_4Cl$ and $ZnCl_2$ fills the gap
- the electrode reactions are complex; approximately:

anode: $Zn(s) \rightarrow Zn^{2+} + 2e^-$

cathode: $MnO_2 + NH_4^+ + e^- \rightarrow MnO(OH) + NH_3$

- Mn is reduced from +4 to +3; the $NH_3$ formed complexes $Zn^{2+}$ as $[Zn(NH_3)_4]^{2+}$; potential nearly 1.5 V

### Mercury cell

- for low-current devices such as hearing aids and watches (Figure 2.I)
- anode: zinc–mercury amalgam; cathode: paste of HgO and carbon; electrolyte: paste of KOH and ZnO

anode: $Zn(Hg) + 2OH^- \rightarrow ZnO(s) + H_2O + 2e^-$

cathode: $HgO + H_2O + 2e^- \rightarrow Hg(l) + 2OH^-$

### Mercury cell: overall

$$Zn(Hg) + HgO(s) \rightarrow ZnO(s) + Hg(l)$$

- cell potential about 1.35 V
- it stays constant for the cell's whole life, because no ion in solution whose concentration could change takes part in the overall reaction

**Figure 2.I** Commonly used mercury cell. The reducing agent is zinc and the oxidising agent is mercury (II) oxide.

- a small button-shaped cell cut open: anode cap on top over the zinc anode, a separator, then the cathode, all held in a cell can sealed by a gasket

## 2.14 Secondary Batteries

### Secondary: rechargeable

- after use, current passed through in the **opposite** direction recharges it for further use
- a good secondary cell survives very many discharge–charge cycles
- the most important: the **lead storage battery** (Figure 2.J), used in vehicles and inverters

### Lead storage battery: construction

- anode: lead
- cathode: a lead grid packed with lead dioxide ($PbO_2$)
- electrolyte: 38% sulphuric acid solution

### Lead storage battery: reactions in use

anode: $Pb(s) + SO_4^{2-}(aq) \rightarrow PbSO_4(s) + 2e^-$

cathode: $PbO_2(s) + SO_4^{2-}(aq) + 4H^+(aq) + 2e^- \rightarrow PbSO_4(s) + 2H_2O(l)$

overall: $Pb(s) + PbO_2(s) + 2H_2SO_4(aq) \rightarrow 2PbSO_4(s) + 2H_2O(l)$

- charging reverses the reaction: $PbSO_4(s)$ on the anode becomes Pb again, and on the cathode becomes $PbO_2$

**Figure 2.J** The Lead storage battery.

- a cut-away case holding alternating plates immersed in 38% sulphuric acid solution, with anode and cathode terminals on top
- negative plates: lead grids filled with spongy lead; positive plates: lead grids filled with $PbO_2$

### Nickel–cadmium cell

- another important secondary cell (Figure 2.K); lasts longer than the lead storage cell but costs more to make
- the working details and electrode reactions for charging and discharging are not covered; the overall discharge reaction is:

$$Cd(s) + 2Ni(OH)_3(s) \rightarrow CdO(s) + 2Ni(OH)_2(s) + H_2O(l)$$

**Figure 2.K** A rechargeable nickel-cadmium cell in a jelly roll arrangement and separated by a layer soaked in moist sodium or potassium hydroxide.

- a positive plate, a separator and a negative plate rolled together into a cylinder inside the cell case

## 2.15 Fuel Cells

### The case against thermal plants

- generating electricity in thermal plants is not very efficient and pollutes heavily
- the combustion heat of fossil fuels (coal, gas, oil) turns water into high-pressure steam, which drives a turbine to make electricity
- a galvanic cell turns chemical energy straight into electricity, with high efficiency

### What a fuel cell is

- cells can now be built in which reactants are fed to the electrodes continuously and products are removed from the electrolyte compartment continuously
- **fuel cells**: galvanic cells designed to convert the combustion energy of fuels such as hydrogen, methane or methanol directly into electrical energy

### The $H_2$–$O_2$ fuel cell

- among the most successful; it reacts hydrogen with oxygen to give water (Figure 2.L)
- powered the Apollo space programme; the water vapour formed was condensed and added to the astronauts' drinking water
- hydrogen and oxygen pass as bubbles through porous carbon electrodes into strong aqueous NaOH; finely divided Pt or Pd in the electrodes speeds up the electrode reactions

cathode: $O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH^-(aq)$

anode: $2H_2(g) + 4OH^-(aq) \rightarrow 4H_2O(l) + 4e^-$

overall: $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$

- runs as long as reactants keep coming; efficiency about 70%

**Figure 2.L** Fuel cell using $H_2$ and $O_2$ produces electricity.

- two porous electrodes with aqueous electrolyte between them: $H_2$ fed in at the anode (–), $O_2$ at the cathode (+)
- water ($H_2O$) leaves from the top of the cell

### Fuel cells: prospects

- about 70% efficient, against about 40% for thermal plants
- huge advances in electrode materials, catalysts and electrolytes are raising efficiency further; fuel cells have been tried experimentally in automobiles
- they cause no pollution, and given their future importance many kinds have been built and tested

## 2.16 Corrosion

### What corrosion is

- a slow coating of metal surfaces with oxides or other salts of the metal
- examples: rusting of iron, tarnishing of silver, the green coating on copper and bronze
- does immense damage to buildings, bridges, ships and all metal objects, iron ones above all; crores of rupees are lost to it each year

### An electrochemical process

- the metal loses electrons to oxygen and forms oxides
- rusting of iron needs both water and air
- the chemistry is complicated but is essentially electrochemical

### Anodic and cathodic spots

- at one spot on the iron (Figure 2.M) oxidation occurs; that spot is the **anode**:

$$2Fe(s) \rightarrow 2Fe^{2+} + 4e^- \quad E^{\ominus}_{(Fe^{2+}/Fe)} = -0.44\ V$$

- the released electrons travel through the metal to another spot, where they reduce oxygen in the presence of $H^+$; that spot is the **cathode**
- the $H^+$ is thought to come from $H_2CO_3$ formed as $CO_2$ from air dissolves in water, and possibly from other dissolved acidic oxides

$$O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l) \quad E^{\ominus}_{H^+|O_2|H_2O} = 1.23\ V$$

$$\text{overall: } 2Fe(s) + O_2(g) + 4H^+(aq) \rightarrow 2Fe^{2+}(aq) + 2H_2O(l) \quad E^{\ominus}_{(cell)} = 1.67\ V$$

- atmospheric oxygen oxidises $Fe^{2+}$ further to $Fe^{3+}$, which appears as rust, hydrated ferric oxide ($Fe_2O_3.xH_2O$), releasing more $H^+$

**Figure 2.M** Corrosion of iron in atmosphere

- a water droplet on an iron surface; within the metal, one region acts as anode and another as cathode
- oxidation: $Fe(s) \rightarrow Fe^{2+}(aq) + 2e^-$
- reduction: $O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$
- atmospheric oxidation: $2Fe^{2+}(aq) + 2H_2O(l) + \frac{1}{2}O_2(g) \rightarrow Fe_2O_3(s) + 4H^+(aq)$

### Preventing corrosion

- vitally important: it saves money and helps avoid accidents such as a bridge collapsing or a key part failing
- simplest: keep the surface from contact with the atmosphere by paint or certain chemicals (e.g. bisphenol), or by coating with other metals (Sn, Zn, etc.) that are inert or react to protect the object
- electrochemical: attach a **sacrificial electrode** of another metal (Mg, Zn, etc.) that corrodes in place of the object

### The Hydrogen Economy

- fossil fuels (coal, oil, gas) are today's main energy source; as more people seek a better standard of living, energy demand will grow; per capita energy use (for productive purposes, not waste) is a measure of development
- $CO_2$ from burning fossil fuels drives the 'Greenhouse Effect', warming the Earth's surface, melting polar ice and raising sea levels
- low-lying coasts will flood and island nations such as the Maldives risk being submerged entirely; carbon-based fuels must be curbed to avert this
- hydrogen is an ideal alternative: burning it yields only water
- it must be produced by splitting water with solar energy, making it a renewable, non-polluting energy source; this is the vision of the **Hydrogen Economy**
- making hydrogen by electrolysis of water and burning it in fuel cells will both matter in future, and both rest on electrochemical principles

### Intext questions

**Q13** Give the chemistry of recharging a lead storage battery, pointing out every material involved in recharging.

**Q14** Name two fuels besides hydrogen that fuel cells can use.

**Q15** Explain how the rusting of iron can be pictured as the setting up of an electrochemical cell.

## Summary

### Cells

- an electrochemical cell has two metal electrodes in electrolytic solution(s); the ionic conductor, or electrolyte, is a key part
- two types: **galvanic**, turning the chemical energy of a spontaneous redox reaction into electrical work; **electrolytic**, using electrical energy to drive a non-spontaneous one

### Potentials

- standard electrode potentials are referred to the hydrogen electrode, taken as zero
- $E^{\ominus}_{(cell)} = E^{\ominus}_{cathode} - E^{\ominus}_{anode}$
- linked to $\Delta_r G^{\ominus} = -nFE^{\ominus}_{(cell)}$ and to $\Delta_r G^{\ominus} = -RT \ln K$; the Nernst equation gives the concentration dependence of electrode and cell potentials

### Conductance

- $\kappa$ of an electrolytic solution depends on electrolyte concentration, the solvent and temperature; $\Lambda_m = \kappa/c$
- on dilution $\kappa$ falls but $\Lambda_m$ rises: slowly for strong electrolytes, very steeply for weak electrolytes in very dilute solution
- Kohlrausch showed that an electrolyte's molar conductivity at infinite dilution adds up from the separate contributions of its ions

## Summary, continued

- the sum rule for ions is the **law of independent migration of ions**, which has many applications
- ions carry the current through the solution, while oxidation and reduction of ions happen at the electrodes
- batteries and fuel cells are very useful kinds of galvanic cell; corrosion of metals is essentially electrochemical; electrochemical principles underpin the Hydrogen Economy

## Exercises

**E1** Put Al, Cu, Fe, Mg and Zn in the order in which they displace one another from solutions of their salts.

**E2** Standard electrode potentials: $K^+/K = -2.93$ V, $Ag^+/Ag = 0.80$ V, $Hg^{2+}/Hg = 0.79$ V, $Mg^{2+}/Mg = -2.37$ V, $Cr^{3+}/Cr = -0.74$ V. List these metals in increasing order of reducing power.

**E3** Draw the galvanic cell for $Zn(s) + 2Ag^+(aq) \rightarrow Zn^{2+}(aq) + 2Ag(s)$ and show: (i) which electrode carries negative charge; (ii) what carries the current in the cell; (iii) the reaction at each electrode.

**E4** Find the standard cell potentials of galvanic cells running: (i) $2Cr(s) + 3Cd^{2+}(aq) \rightarrow 2Cr^{3+}(aq) + 3Cd$; (ii) $Fe^{2+}(aq) + Ag^+(aq) \rightarrow Fe^{3+}(aq) + Ag(s)$. Also find $\Delta_r G^\ominus$ and the equilibrium constant of each.

**E5** Give the Nernst equation and the emf at 298 K of each cell: (i) $Mg(s)|Mg^{2+}(0.001M)||Cu^{2+}(0.0001\ M)|Cu(s)$; (ii) $Fe(s)|Fe^{2+}(0.001M)||H^+(1M)|H_2(g)(1bar)|Pt(s)$; (iii) $Sn(s)|Sn^{2+}(0.050\ M)||H^+(0.020\ M)|H_2(g)(1\ bar)|Pt(s)$; (iv) $Pt(s)|Br^-(0.010\ M)|Br_2(l)||H^+(0.030\ M)|H_2(g)(1\ bar)|Pt(s)$.

**E6** Button cells, common in watches and other devices, run the reaction $Zn(s) + Ag_2O(s) + H_2O(l) \rightarrow Zn^{2+}(aq) + 2Ag(s) + 2OH^-(aq)$. Find $\Delta_r G^\ominus$ and $E^\ominus$ for it.

**E7** Define conductivity and molar conductivity of an electrolyte solution, and discuss how each varies with concentration.

**E8** 0.20 M KCl at 298 K has a conductivity of 0.0248 S $cm^{-1}$. What is its molar conductivity?

**E9** A conductivity cell filled with 0.001M KCl at 298 K has a resistance of 1500 Ω. The conductivity of 0.001M KCl at 298 K is $0.146 \times 10^{-3}$ S $cm^{-1}$. Find the cell constant.

## Answers to some intext questions

| Question | Answer |
|---|---|
| Q5 | $E_{(cell)} = 0.91$ V |
| Q6 | $\Delta_r G^{\ominus} = -45.54$ kJ mol$^{-1}$, $K_c = 9.62 \times 10^7$ |
| Q9 | 0.114, $3.67 \times 10^{-4}$ mol L$^{-1}$ |

## Exercises continued

**E10** The conductivity of sodium chloride at 298 K was measured at several concentrations:

| Concentration / M | 0.001 | 0.010 | 0.020 | 0.050 | 0.100 |
|---|---|---|---|---|---|
| $10^2 \times \kappa$ / S m$^{-1}$ | 1.237 | 11.85 | 23.15 | 55.53 | 106.74 |

Work out $\Lambda_m$ at each concentration, plot $\Lambda_m$ against $c^{1/2}$, and find $\Lambda_m^0$.

**E11** 0.00241 M acetic acid has a conductivity of $7.896 \times 10^{-5}$ S cm$^{-1}$. Find its molar conductivity, and, taking $\Lambda_m^0$ for acetic acid as 390.5 S cm$^2$ mol$^{-1}$, its dissociation constant.

**E12** What charge is needed to reduce (i) 1 mol of $Al^{3+}$ to Al; (ii) 1 mol of $Cu^{2+}$ to Cu; (iii) 1 mol of $MnO_4^-$ to $Mn^{2+}$?

**E13** How many faradays are needed to obtain (i) 20.0 g of Ca from molten $CaCl_2$; (ii) 40.0 g of Al from molten $Al_2O_3$?

**E14** How many coulombs are needed to oxidise (i) 1 mol of $H_2O$ to $O_2$; (ii) 1 mol of FeO to $Fe_2O_3$?

**E15** $Ni(NO_3)_2$ solution is electrolysed between platinum electrodes at 5 amperes for 20 minutes. What mass of Ni is deposited on the cathode?

**E16** Electrolytic cells A, B and C, holding $ZnSO_4$, $AgNO_3$ and $CuSO_4$ solutions, are joined in series. A steady 1.5 amperes flows until 1.45 g of silver has deposited on the cathode of cell B. For how long did the current flow, and what masses of copper and zinc were deposited?

**E17** From the standard electrode potentials in Table 2.A, decide whether reaction is feasible between: (i) $Fe^{3+}$(aq) and $I^-$(aq); (ii) $Ag^+$(aq) and Cu(s); (iii) $Fe^{3+}$(aq) and $Br^-$(aq); (iv) Ag(s) and $Fe^{3+}$(aq); (v) $Br_2$(aq) and $Fe^{2+}$(aq).

**E18** Predict the products of electrolysis of: (i) aqueous $AgNO_3$ with silver electrodes; (ii) aqueous $AgNO_3$ with platinum electrodes; (iii) dilute $H_2SO_4$ with platinum electrodes; (iv) aqueous $CuCl_2$ with platinum electrodes.